Kangaroo arms and the BeyondMimic action scale

Unitree G1 vs PAL Kangaroo. Summary of the SuperMimic gain A/B and the 7-DoF WBC wrist report (M. Actis).
pal_mjlab, October 2026.

1 The rule

BeyondMimic, and SuperMimic after it, derive gains and action scale per joint from the motor alone:

$$K_p = I_a\,\omega^2, \quad K_d = 2\zeta I_a\,\omega, \quad \omega = 2\pi f,\ \zeta = 2, \qquad q_{sp} = q_0 + \alpha\,a, \quad \alpha = c\,\frac{\tau_{\max}}{K_p},\ c = 0.25.$$

I_a is the armature (reflected motor inertia), $\tau_{\max}$ the effort limit, q_0 the default pose, a the network output and α the setpoint offset per action unit (rad/unit). mjlab sends q_{sp}, minus the calibration offset, to a MuJoCo position actuator: $\tau = \mathrm{clip}\big(K_p(q_{sp} - q) - K_d\dot{q},\ \pm\tau_{\max}\big)$. The policy only moves the setpoint; the one spring pulls the joint toward it.

2 What an action unit means

The same law in action units, using $q_{sp} = q_0 + \alpha a$: $\tau = \underbrace{c\,\tau_{\max}\,a}_{\text{action}} - \underbrace{K_p(q - q_0)}_{\text{where } q \text{ sits vs } q_0} - \underbrace{K_d\,\dot{q}}_{\text{damping}}$. This is bookkeeping, not a second spring: actions are measured from q_0, so a joint resting at q (setpoint on the joint, $\tau \approx 0$) needs $a = (q - q_0)/\alpha$.

- **Torque per unit:** $\partial\tau/\partial a = c\,\tau_{\max}$, i.e. 25% of the effort limit, at a fixed state only.
- **Holding** away from q_0 is a large action with almost no torque, so $|a| > 4$ is not saturation. WBC example, arm joint 1 at $a = 4.5$: $+56\,\mathrm{Nm}$ action term $- 49\,\mathrm{Nm}$ $(q - q_0)$ term $= 7\,\mathrm{Nm}$ net.
- **Moving** takes $\Delta a \approx \dot{q}\,\Delta t/\alpha$ per step. **Exploring** with noise σ moves the setpoint by $\sigma\alpha$.

So $K_p/\tau_{\max} = c/\alpha$ is the **exchange rate between joint motion and action units**. Torques don't depend on it, but the action-rate penalty, the exploration noise and the output range do. It follows from the hardware and the frequency:

$$\frac{K_p}{\tau_{\max}} = \frac{I_a}{\tau_{\max}}\,\omega^2 \quad \Rightarrow \quad \frac{(K_p/\tau_{\max})_{\text{Kangaroo}}}{(K_p/\tau_{\max})_{\text{G1}}} = \frac{(I_a/\tau_{\max})_{\text{Kangaroo}}}{(I_a/\tau_{\max})_{\text{G1}}} \cdot \Big(\frac{f_{\text{Kangaroo}}}{f_{\text{G1}}}\Big)^2.$$

3 Unitree G1 vs Kangaroo

Joints	Motor	I_a ($\mathrm{kg\,m^2}$)	$\tau_{\max}$ (Nm)	f (Hz)	K_p (Nm/rad)	$I_a/\tau_{\max}$ ($10^{-3}\,\mathrm{s^2}$)	$K_p/\tau_{\max}$ (1/rad)	α (rad/unit)
Unitree G1, BeyondMimic (mjlab)								
Shoulders, elbow, wrist roll	5020	0.0036	25	10	14.3	0.14	0.57	0.44
Hip pitch/yaw, waist yaw	7520-14	0.0102	88	10	40.2	0.12	0.46	0.55
Hip roll, knee	7520-22	0.0251	139	10	99.1	0.18	0.71	0.35
Waist pitch/roll, ankles	2×5020	0.0072	50	10	28.5	0.14	0.57	0.44
Wrist pitch/yaw	4010	0.0043	5	10	16.8	0.85	3.36	0.075
PAL Kangaroo (PR #126 constants)								
Leg 1–3 (hip)		0.20–0.50	80–230	3.5	97–242	1.4–3.1	0.70–1.51	0.17–0.36
Leg 4, 5 (ankle)		0.05–0.10	82–140	3.5	24–48	0.6–0.7	0.29–0.35	0.72–0.85
Arm 1, 2 + waist	S+, gear 121	0.483	50	3.5	233.7	9.7	**4.67**	**0.053**
Arm 3–5	S−, gear 101	0.305	25	3.5	147.5	12.2	**5.90**	**0.042**
Arm 6, 7	XS, gear 101	0.203	25	3.5	98.2	8.1	**3.93**	**0.064**
Arms + waist at 1 Hz	as above	as above	as above	1	8.0–19.1	as above	0.32–0.48	0.52–0.78

Arm vs arm. Kangaroo's geared arm motors have 56–134× G1's armature for only 1–2× its effort, so $I_a/\tau_{\max}$ is **56–85**× larger. At 3.5 Hz instead of 10 Hz, ω^2 shrinks by 8.2×, so $K_p/\tau_{\max}$ ends up **7–10**× G1's: 0.57 on G1 against 3.9–5.9 on Kangaroo. Matching G1 needs $f \approx 10/\sqrt{56\text{–}85} \approx 1.1\text{–}1.3$ Hz. Kangaroo's legs are already close to G1 (0.29–1.51). The only G1 joints in Kangaroo's arm regime are the two 5 Nm wrist motors (3.36), whose motion barely changes the tracked bodies. On Kangaroo the regime covers 14 arm joints plus the waist.

4 What it does to learning

- **Wrist exploration (WBC report).** At $\alpha \approx 0.04$–0.06, exploration barely moves joints 5–7. Their effect, end-effector orientation, has a reward flat near 0, so PPO gets no signal and leaves the wrist at q_0.
- **Action size and action-rate penalty (SuperMimic).** The arm references need $|a| = 7.8$ (median) to 40.8 (p95). The penalty $-0.1\|\Delta a\|^2$ scales with $(K_p/\tau_{\max})^2$ for the same motion: 28% of the maximum per-step reward at p95 at 3.5 Hz, against 0.2% at 1 Hz.
- **Not the plant.** Torque saturation is near zero in both studies.

5 Evidence

Study	Change	Result
WBC, 7-DoF (M. Actis)	3.5 Hz gains kept; α on joints 5–7 only, $\times 4$ (c=1) and $\times 16$ (c=4)	Orientation reward rises close to the 1 Hz run; position and root unchanged. The scale alone is the main factor for the wrist.
SuperMimic PPO-Sim, 1296 clips	Arms + waist at 1 Hz vs 3.5 Hz (gains and α together), 1000 iterations	Arm error 21.2→15.3 cm (better on 90% of clips), legs 11.3→8.6 cm (95%), tracking return 0.53→0.62, saturation 0.8%→0.04%. At 3.5 Hz the arm error grew during training.

1 Hz multiplies α by 12.25, the same as $c \approx 3.06$ at 3.5 Hz, which lies within the WBC report's recommended $c = 3$–4 for the wrist.

6 Options and rule of thumb

1. **Arms and waist at ∼1 Hz** (current SuperMimic baseline). Fixes α, but the servos get 12× less stiff. Gravity sag τ_g/K_p grows 12×, and the lag $(K_d/K_p)\dot{q} = (2\zeta/\omega)\dot{q}$ goes from 0.18 to 0.64 s·$\dot{q}$. Hardware transfer is open.
2. **Keep the gains, set α per joint** ($c \approx 3$–4 on the arms). The plant is untouched and α matches the 1 Hz run. Tested on the wrist only.
3. **Measure actions from another anchor**, which removes the $(q - q_0)$ term. Either the reference, $q_{sp} = q_{\text{ref}} + \alpha a$, or the current position, $q_{sp} = q + \alpha a$ (mjlab's `RelativeJointPositionAction`; that one loses the position anchor). Untested; both change the paper's action definition.

Rule of thumb: when moving BeyondMimic or SuperMimic to a new robot, check $K_p/\tau_{\max}$ per joint and aim for G1's ≈ 0.5–0.7 per rad ($\alpha \approx 0.35$–0.55 rad). The frequency alone doesn't tell you this. **Limits:** single seed per run; the 1 Hz runs change plant and scale together; the reward gradient was not measured directly; simulation only.